

**K L Deemed to be University***(Koneru Lakshmaiah Education Foundation)*

GREEN FIELDS, VADDESWAREM · TERM 3, AY 2025-2026 · PROJECT-BASED LEARNING

TERM-END EXAMINATION · PAPER SET 1

COURSE	COMPUTATIONAL FOUNDATIONS FOR ARTIFICIAL INTELLIGENCE
CODE	25SC1306E
EXAMINATION	Term-End Examination
PAPER	Set 1
DURATION	180 minutes
MAX. MARKS	90
COORDINATOR	Vidyullatha Pellakuri
MODE	Project-Based Learning

INSTRUCTIONS TO CANDIDATES

- Answer ALL six questions. Q1–Q3 carry 5 marks each (short conceptual). Q4–Q6 carry 25 marks each (long scenario / case-study).
- Each question covers one Course Outcome. Sub-parts within a question must be answered in order.
- Show all working, intermediate steps, code listings, diagrams, derivations and reasoning traces.
- Time allowed: 180 minutes. Maximum marks: 90.
- Calculator allowed. No mobile phones, smart watches, or AI assistants.

Q1.**C01**

Module 1: Rapido bike-taxi agent — PEAS

[N/A Marks]**SCENARIO / CASE STUDY**

Rapido bike-taxi rider in Hyderabad. A Rapido rider operates a bike-taxi service through the Rapido app. Riders accept ride requests, navigate to passenger pickup, deliver them to drop locations, and collect cashless payment. Today rider R47 is at HiTec City junction with 65% phone-battery, ₹0 cash on hand (UPI-only).

SUB	QUESTION	MARKS	CO	BTL
(a)	State the PEAS specification for this Rapido bike-taxi agent. Provide all 4 components in a 4-row table: Performance (3 metrics minimum), Environment (6 dimensions classified), Actuators (3+), Sensors (3+).	5	C01	BTL3

Q2.**C02**

Module 2: Search algorithm comparison

[N/A Marks]**SCENARIO / CASE STUDY**

Search-algorithm property comparison. Different search algorithms have different completeness, optimality, time, and space properties. Practitioners must pick the right algorithm for the problem at hand.

SUB	QUESTION	MARKS	CO	BTL
(a)	Compare BFS, DFS, UCS, Greedy Best-First, A* on these properties: <i>completeness, optimality (when h is admissible/consistent), time complexity, space complexity</i> . Present as a 5-row $\times$ 4-column table. State one practical scenario where each algorithm is the best choice.	5	C02	BTL4

Q3.

C03

Module 3: Constraint propagation concepts

[N/A Marks]

SCENARIO / CASE STUDY

Constraint propagation in CSPs. CSPs use constraint propagation to reduce variable domains BEFORE backtracking begins. Propagation algorithms differ in propagation power.

SUB	QUESTION	MARKS	CO	BTL
(a)	Define and contrast the three constraint-propagation techniques: (i) <i>node consistency</i> , (ii) <i>arc consistency (AC-3)</i> , (iii) <i>path consistency</i> . For each: state what it propagates, its time complexity, and when it's worth running. Give a 1-line example showing how AC-3 prunes more than node consistency.	5	C03	BTL4

Q4.

C04

Module 4: IPL Auction Bid Strategy — Utility, Minimax, Expectimax

[N/A Marks]

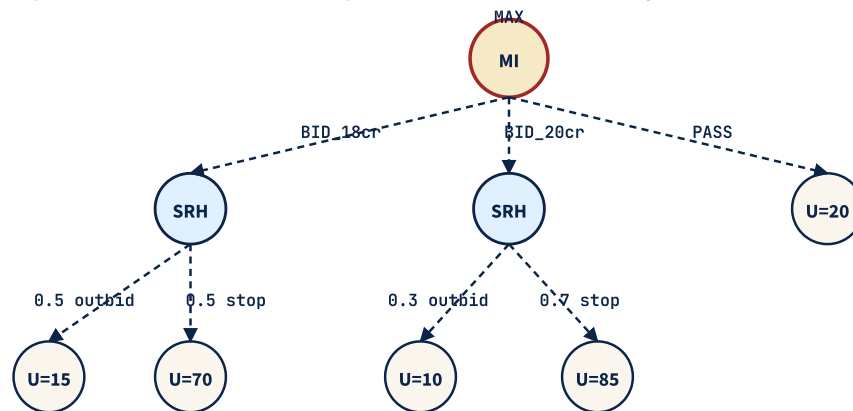
SCENARIO / CASE STUDY

IPL 2026 mega-auction bid strategy — utility theory + minimax + expectimax. Mumbai Indians (MI) team owners must decide their **bidding strategy for player Pat Cummins** (base price ₹2 cr, current bid ₹17 cr from Sunrisers Hyderabad, MI's purse ₹35 cr remaining, 3 marquee slots open). MI's analytics team has built a 2-stage decision tree:

- **MI's first move:** BID_18cr / BID_20cr / PASS
- **SRH's response** (random — modelled as expectimax with prob 0.5 outbid, 0.5 stop): each branch leads to a leaf with a quantified utility (expected wins-above-replacement × revenue impact).

The decision tree leaves contain utility values in 'utility units' where 100 utility $\approx$ 1 IPL trophy $\approx$ ₹100 cr revenue impact. MI's owner must pick a strategy maximising expected utility, treating SRH's bid as a chance node.

IPL auction expectimax · MI-bid (MAX) → SRH-response (CHANCE) → outcome utility



Leaf utilities: BID_18cr branch: outbid→15, stop→70 · BID_20cr branch: outbid→10, stop→85 · PASS: 20

SRH probabilities differ: more aggressive at 18cr (0.5 outbid), conservative at 20cr (0.3 outbid).

SUB	QUESTION	MARKS	CO	BTL
(a)	Define rational agency under uncertainty. State the <i>maximum expected utility (MEU) principle</i> as the foundational rationality criterion for decision-making under uncertainty. Distinguish between (i) <i>utility</i> and <i>preference</i> , (ii) <i>expected utility</i> vs <i>certainty equivalent</i> , (iii) <i>risk-neutral</i> , <i>risk-averse</i> , and <i>risk-seeking</i> agents and the shape of their utility functions. State why the MEU principle is normative (what an agent ought to do) but not always descriptive (real humans deviate via Allais paradox, Kahneman-Tversky prospect theory).	5	C04	BTL4
(b)	Expectimax computation on the IPL tree. Compute the expected utility at each chance node by $\sum p \times \text{utility}$: • $EV(\text{BID_18cr}) = 0.5 \times 15 + 0.5 \times 70 = ?$	8	C04	BTL5

SUB	QUESTION	MARKS	CO	BTL
	• $EV(\text{BID_20cr}) = 0.3 \times 10 + 0.7 \times 85 = ?$ • $U(\text{PASS}) = 20$ (deterministic leaf) Then pick the MAX among the 3 actions. State (i) the chosen MI bid strategy, (ii) the expected utility of that strategy, (iii) what changes if MI is risk-averse (utility function $u(\text{money}) = \log(\text{money} + 100)$ instead of linear).			
(c)	Minimax vs Expectimax — when each is appropriate. The IPL tree above uses <i>expectimax</i> because SRH's bid is uncertain (probabilistic). Suppose instead MI faced an adversarial 'auction agent' acting to MINIMISE MI's utility (e.g., a competing team owner deliberately bidding up to drain MI's purse). State (i) what changes in the tree evaluation under minimax (replace Σ with min), (ii) compute the minimax value for the BID_18cr and BID_20cr branches, (iii) discuss when alpha-beta pruning would help. (iv) Which model — expectimax or minimax — is more realistic for IPL auctions, and why?	8	C04	BTL5
(d)	Bounded rationality. The MI owner has 30 seconds to make the bid decision. They cannot enumerate the full game-tree (8 future bid rounds, 200+ players, ₹100 cr+ branching). State the practical adaptations: (i) <i>iterative deepening</i> applied to game-tree search; (ii) <i>evaluation functions</i> that approximate true utility at non-terminal nodes; (iii) <i>quiescence search</i> for highly-volatile sub-trees (auction wars). Cite the connection to Stockfish (chess) and AlphaZero (Go) — they ALL use bounded-rationality variants of minimax/expectimax + neural-net evaluators.	4	C04	BTL5

Q5.

C05

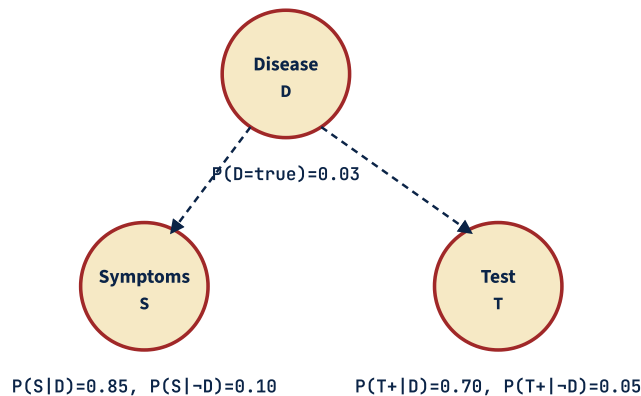
Module 5: COVID Spike-protein Bayesian Disease Screening

[N/A Marks]

SCENARIO / CASE STUDY

COVID rapid antigen test interpretation — Bayes' theorem + Bayesian network. A Pune diagnostic lab uses RT-PCR (gold standard, slow) and rapid-antigen-test (RAT, quick but less accurate). The clinical context: in May 2026, Pune has community prevalence of COVID-Spike-protein-strain at ~3%. The RAT has sensitivity 70% ($P(\text{test+} \mid \text{disease}) = 0.70$) and specificity 95% ($P(\text{test-} \mid \text{no disease}) = 0.95$). A patient walks in with symptoms (cough + fever) and tests POSITIVE. The clinician must decide: trust the test, or order RT-PCR confirmation?

Bayesian network for COVID screening • D=Disease, S=Symptoms, T=Test



Joint factorisation:

$$P(D, S, T) = P(D) \cdot P(S|D) \cdot P(T|D)$$

Conditional independence:

$$S \perp T \mid D$$

Today's evidence:

S = true (cough + fever)

T = positive

Prior $P(D) = 3\%$ (Pune base rate)

SUB QUESTION

MARKS

C0

BTL

- (a) **Bayes' theorem on RAT alone (no symptoms).** Compute $P(\text{Disease} \mid \text{Test}=+)$ using Bayes' theorem. $P(\text{Disease}) = 0.03$; $P(T+ \mid D) = 0.70$; $P(T+ \mid \neg D) = 0.05$. **Show all steps:** (i) compute $P(T+) = P(T+|D)P(D) + P(T+|\neg D)P(\neg D)$; (ii) apply Bayes: $P(D|T+) = P(T+|D)P(D)/P(T+)$. State the answer in a sentence: 'A positive RAT in Pune means the patient has X% probability of being infected.'

5

C05

BTL3

- (b) **Bayesian inference with symptoms + test combined.** Now add the symptom evidence $S=\text{true}$. Use the network's conditional-independence assumption ($S \perp T \mid D$). The joint probability factors as $P(D, S, T) = P(D) \cdot P(S|D) \cdot P(T|D)$. Compute $P(\text{Disease} \mid S=\text{true}, T=+)$ by:

8

C05

BTL5

- $P(D, S=T, T=+) = 0.03 \times 0.85 \times 0.70 = ?$
- $P(\neg D, S=T, T=+) = 0.97 \times 0.10 \times 0.05 = ?$
- $P(D \mid S=T, T=+) = \text{numerator} / (\text{num} + \text{denom}) = ?$

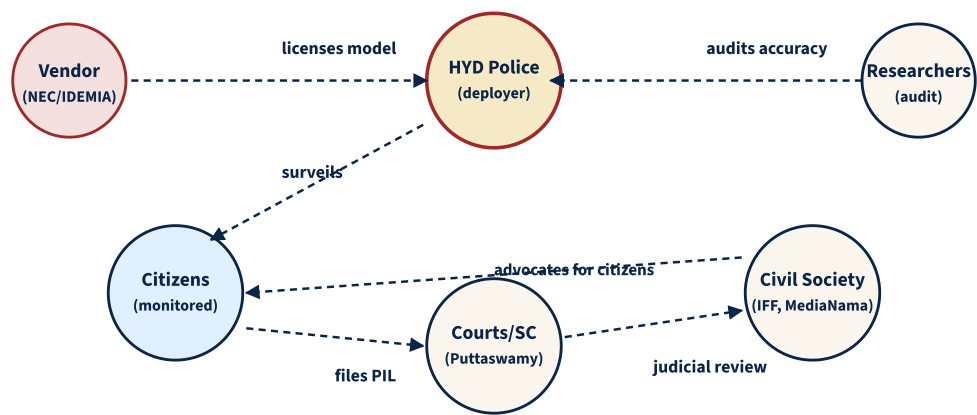
State the result. Compare with the RAT-only answer from part (a). State why combined evidence INCREASES the posterior probability above 30%.

SUB	QUESTION	MARKS	CO	BTL																									
(c)	Markov chains and disease progression. A patient's disease state evolves over 7 days as a Markov chain: states = {Healthy (H), Infected-asymptomatic (I_A), Infected-symptomatic (I_S), Recovered (R)}. Transition matrix: <table> <tr> <th>From \ To</th><th>H</th><th>I_A</th><th>I_S</th><th>R</th></tr> <tr> <th>H</th><td>0.97</td><td>0.03</td><td>0</td><td>0</td></tr> <tr> <th>I_A</th><td>0</td><td>0.40</td><td>0.50</td><td>0.10</td></tr> <tr> <th>I_S</th><td>0</td><td>0</td><td>0.30</td><td>0.70</td></tr> <tr> <th>R</th><td>0</td><td>0</td><td>0</td><td>1.00</td></tr> </table> (i) Compute the state distribution after Day 2 starting from Day 0 state = (1, 0, 0, 0) (healthy). Show matrix multiplication. (ii) State whether {H, I_A, I_S, R} is an absorbing Markov chain. Identify the absorbing state. (iii) Compute the expected number of days until absorption from initial state I_A.	From \ To	H	I_A	I_S	R	H	0.97	0.03	0	0	I_A	0	0.40	0.50	0.10	I_S	0	0	0.30	0.70	R	0	0	0	1.00	8	C05	BTL5
From \ To	H	I_A	I_S	R																									
H	0.97	0.03	0	0																									
I_A	0	0.40	0.50	0.10																									
I_S	0	0	0.30	0.70																									
R	0	0	0	1.00																									
(d)	HMM for symptom progression. The clinician observes symptoms (mild/moderate/severe) but the underlying disease state is hidden. State the HMM 5-tuple: (states, observations, initial-distribution, transition-matrix, emission-matrix). Cite the three classical HMM tasks: (i) <i>Forward algorithm</i> — likelihood of observation sequence; (ii) <i>Viterbi</i> — most-likely state sequence; (iii) <i>Baum-Welch</i> — parameter learning. State which task the clinician needs (Viterbi: 'what disease state explains the observed symptom progression?').	4	C05	BTL4																									

SCENARIO / CASE STUDY

Hyderabad Police facial-recognition CCTV — responsible AI deployment. Hyderabad City Police is rolling out a facial-recognition system across 500 CCTV cameras (Banjara Hills, Jubilee Hills, Begumpet, HiTec City). The system claims to: (i) identify wanted criminals from a 50 000-face watchlist; (ii) flag 'suspicious behaviour' (loitering > 5 min in commercial zones); (iii) alert officers in real-time. Public concerns: (a) a Telangana High Court PIL challenges the system as violating the Puttaswamy fundamental-right-to-privacy ruling; (b) the watchlist includes 12 000 individuals without active charges (Indian peaceful-protest organisers, journalists); (c) face-recognition models are documented to have higher false-positive rates on darker-skinned faces (Buolamwini & Gebru 2018, Gender Shades). The system is being deployed despite these concerns under the 'public safety' rationale.

Stakeholder map: HYD facial-recognition deployment • 5 stakeholder groups + interaction edges



SUB	QUESTION	MARKS	CO	BTL
(a)	Stakeholder map + responsibilities. List the 6 stakeholders (police, citizens, courts, civil-society, vendor, researchers). For each, state: (i) what value/interest they bring or seek to protect, (ii) their power and influence over the deployment, (iii) what specific action they should take to ensure responsible AI. (e.g., civil-society should publish independent accuracy audits.)	5	C06	BTL4
(b)	Bias evaluation + mitigation. The Buolamwini-Gebru study found commercial face-recognition systems have 1-2% error rates on white males, but 30%+ on darker-skinned females. (i) Define <i>fairness metrics</i> : demographic parity, equalised odds, calibration. (ii) Compute the error-rate-disparity (the ratio of false-positive rate on demographic A vs demographic B) for the worst-case scenario above. (iii) State 3 mitigation strategies (data augmentation with diverse faces, fairness-aware	8	C06	BTL5

SUB	QUESTION	MARKS	CO	BTL
	training, post-hoc threshold calibration). (iv) Discuss why a 'fair' system can still cause harm if deployed without consent — fairness ≠ ethics.			
(c)	Privacy framework: PDP Bill, GDPR, Puttaswamy. India's Digital Personal Data Protection Act 2023 (DPDPA) and the Puttaswamy ruling establish privacy as a fundamental right. State (i) the DPDPA's data-minimisation principle and how it applies to a 50 000-person watchlist with 12 000 non-criminal entries; (ii) the GDPR's Article 22 (right not to be subject to automated decision-making) — does India have an equivalent? (iii) the Puttaswamy three-fold test (legality, legitimate aim, proportionality) and apply it to the facial-recognition deployment. State whether the deployment passes the proportionality test.	8	C06	BTL5
(d)	Engineer's professional responsibility. You are the lead AI engineer at the vendor. The Hyderabad Police asks you to (i) lower the false-negative threshold to 'catch more criminals,' (ii) integrate the system with Aadhaar database for cross-referencing. State your professional response. Cite the IEEE Code of Ethics and ACM Code of Ethics on: respecting privacy, avoiding harm, providing accurate evaluations to stakeholders. State the appropriate escalation path (refuse → escalate to vendor leadership → notify regulator → public-interest disclosure if necessary).	4	C06	BTL5